

Wikipedia FineMath-3+ StarCoder

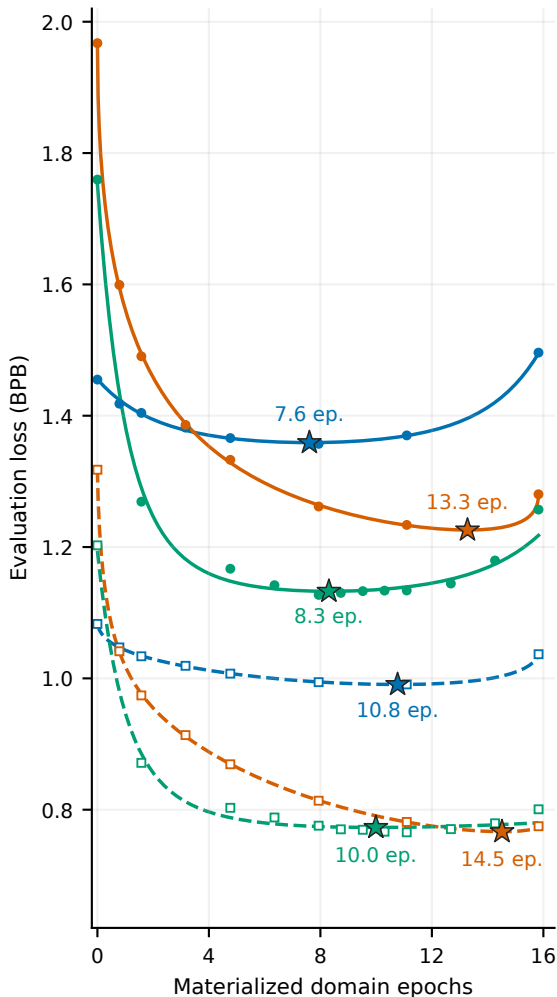
Simulated-epoching proxy ( $2.49 \times 10^{16}$  FLOPs) Target ( $6.66 \times 10^{18}$  FLOPs)

★ Fitted minimum ♦ Fixed epoch count ♦ Simulated-epoching selection

## Different benchmarks

Wikipedia: Wikipedia English | FineMath: MATH-500

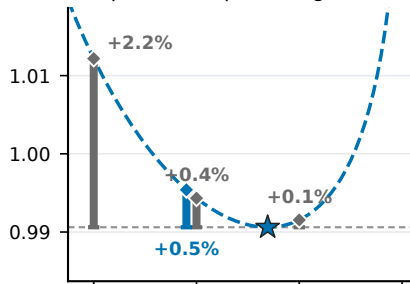
StarCoder: Paloma programming languages



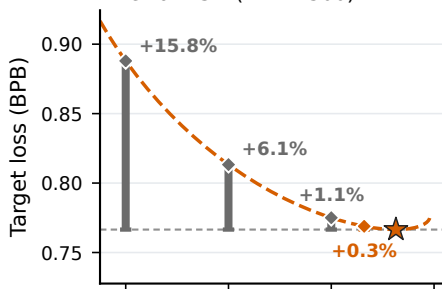
## Fixed epoch count vs. simulated epoching

Excess target loss over the fitted minimum

Wikipedia (Wikipedia English)



FineMath-3+ (MATH-500)



StarCoder (Paloma code)

